ISAAC NASON
Senior Edge & Embedded Software Engineer
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SUMMARY

Senior Software Engineer (Python/C#/.NET) with 15 years of edge compute, full-stack architecture, and telemetry experience, grounded by structural EMT-B and wildland firefighting experience. Specialized in bridging high-throughput physical sensor data (EO/IR, LiDAR, RTSP/WebRTC Video) with containerized edge AI pipelines and React/TypeScript interfaces for rugged environments. I build out complex system architectures from raw CAD concepts to functional deployment.

SKILLS

- **Edge AI & Vision:** Python, OpenCV, Cognex VisionPro, TFLite (Edge AI), Roboflow, MicroPython, CircuitPython
- **Full-Stack & Enterprise:** C#, .NET, WPF, XAML, React, TypeScript, FastAPI, REST APIs, WebSockets, MVC
- **Data, Cloud & Infrastructure:** SQL Server (SSMS), PostgreSQL, Docker, Docker Compose, Linux (Debian), CI/CD, Git
- **Video, Hardware & Telemetry:** WebRTC (WHEP), RTSP, MediaMTX, MAVLink, MQTT, I2C, SPI, UART, ESP32, RP2040, Raspberry Pi, Pixhawk
- **Sensors & Fabrication:** FLIR Lepton 3.5 (Thermal), LiDAR, BNO085/BNO055 IMUs, multi-axis targeting, Autodesk Fusion, Blender, 3D Printing (Bambu Lab X1-Carbon)

EXPERIENCE

NasonNation Robotics | Embedded & Full-Stack Systems Engineer | Feb 2015 – Present

- **NasonNation Command - Distributed Edge/GPU Computer Vision & Biometric Security Platform:**
    - Hybrid Edge-to-GPU Inference Architecture: Engineered a containerized multi-stream RTSP/WebRTC (MediaMTX, FastAPI, React, PostgreSQL) surveillance engine running continuous YOLOv8s ONNX object detection, YuNet 5-point facial landmark alignment, and SFace 128-D biometric vector matching on an edge Linux node with asynchronous Qwen2.5-VL vision-language verification offloaded over LAN to an NVIDIA RTX 3060 Ti.
    - Kinematic Tracking & False-Alarm Suppression: Designed a 2-pass kinematic displacement tracker with bounding-box IoU stationary footprint locking and 5-anchor polygon ray-casting, eliminating stationary vehicle/object false positives and cross-track teleportation.
    - Zero-PII Threat Intelligence & Biometric Pipeline: Built an automated background ingestion daemon integrating the federal FBI Wanted REST API and state fugitive feeds, computing normalized 128-D facial embeddings purely in RAM (zero disk image

retention) for sub-millisecond cosine-similarity watchlist matching alongside auto-clustered visitor dossiers.

- **AeroTherm UAV (Thermal & LiDAR Edge Pipeline):** Designed and fabricated a custom 7-inch autonomous UAV payload integrating a FLIR Lepton 3.5 and LiDAR arrays. Engineered the Python backend for dynamic, real-time thermal targeting running on a headless Raspberry Pi edge pipeline communicating over MAVLink.
- **Aegis Control Sentry Turret:** Architected an AI-powered physical sentry turret utilizing Roboflow and TFLite for local edge perception, real-time targeting, and mechanical tracking. Developed a Flask web interface for live telemetry monitoring.
- **Autonomous UAV Fleet:** Engineered custom multi-rotor UAVs utilizing Python-driven Finite State Machines (FSMs) and MAVLink telemetry for autonomous, GPS-denied indoor navigation.
- **Exes Distributed Exoskeleton:** Built a voice-activated prototype utilizing local edge-compute to drive 550KG servos, integrating Picovoice for offline acoustic processing, alongside a distributed MQTT telemetry network and isolated power architecture.

City of Beaverton | Business Analyst Programmer | May 2021 – Feb 2026

- Architected enterprise software systems and engineered multi-layer APIs using secure tokenization to handle distributed data access.
- Engineered custom UI bypasses for strict manual entry limitations, streamlining data pipelines and eliminating 3 hours of redundant data entry per week.
- Built and optimized mission-critical applications supporting thousands of daily users, operating autonomously to resolve complex bottlenecks.

MyAppMobile, LLC | Senior Software Engineer & Founder | Feb 2015 – May 2021

- Architected and deployed 9 cross-platform edge-device applications (Android/iOS) utilizing .NET, Xamarin, and Google Firebase.
- Refactored legacy codebases and integrated secure APIs to scale mobile solutions handling 800+ concurrent users with 99.9% uptime.

Tektronix | Service Program Specialist (Developer) | Jan 2011 – May 2021

- Served as Technical Lead on a Cognex VisionPro machine vision project to automatically identify defective hardware components from field technician imagery.
- Architected the award-winning "Tektronix Toolbox" C#/WPF desktop application, providing customizable desktop widgets with dynamic UI state views across the enterprise.
- Engineered high-impact internal tools utilizing the Microsoft Power Platform and SSMS to automate worldwide customer support operations, saving colleagues an estimated 10 hours a week on manual data entry.

Forest Grove Fire & Rescue / Oregon Department of Forestry | Firefighter / EMT-B | Jan 2011

- Executed tactical fire suppression and emergency medical response in high-stress environments.
- Utilized operational background to directly inform the ruggedization and deployment strategies for autonomous systems designed to operate in off-grid and hazardous conditions.

EDUCATION & CERTIFICATIONS

- **B.S. Business Management**, Portland State University
- **Certifications:** Criminal Justice Information Services (CJIS) Certified